www.tntextbooks.in

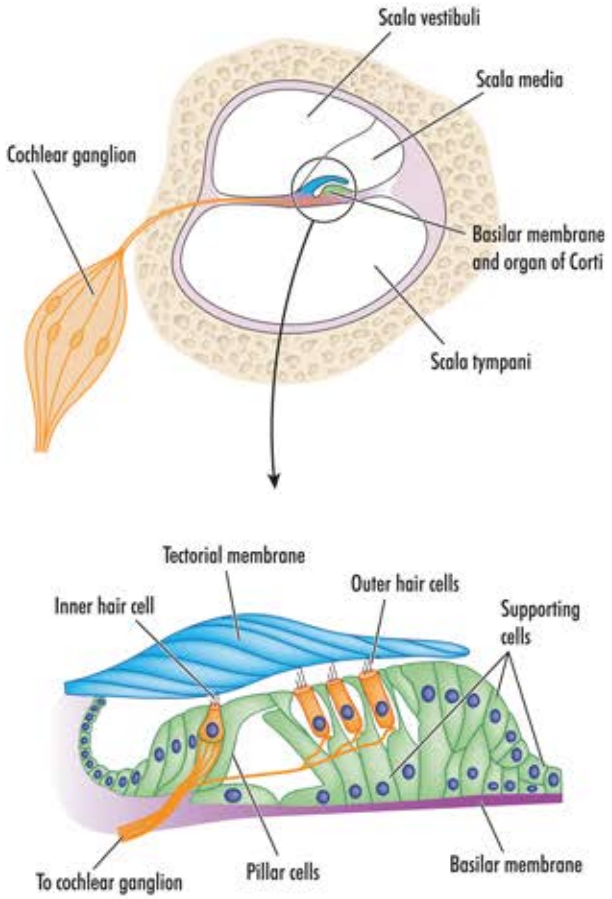


Figure 10.19 Organ of corti

from the external ear by the eardrum and from the internal ear by a thin bony partition; the bony partition contains two small membrane covered openings called the oval window and the round window.

The middle ear contains three ossicles: **malleus** [hammer bone], **incus** [anvil bone] and **stapes** [stirrup bone] which are attached to one another. The malleus is attached to the tympanic membrane and its head articulates with the incus which is the intermediate bone lying between the malleus and stapes. The stapes is attached to the oval window in the inner ear. The ear ossicles transmit sound waves to the inner ear. A tube called Eustachian tube connects the middle ear cavity with the pharynx. This tube helps in equalizing the pressure of air on either sides of the ear drum.

Inner ear is the fluid filled cavity consisting of two parts, the bony labyrinth and the membranous labyrinths. The bony labyrinth consists of three areas: **cochlea, vestibule** and **semicircular canals.** The **cochlea** is a coiled portion consisting of 3 chambers namely: **scala vestibuli** and **scala tympani**- these two are filled with **perilymph;** and the **scala media** is filled with **endolymph.** At the base of the cochlea, the scala vestibule ends at the 'oval window' whereas the scala tympani ends at the 'round window' of the middle ear. The chambers scala vestibuli and scala media are separated by a membrane called Reisner's membrane whereas the scala media and scala tympani are separated by a membrane called **Basilar membrane** (Figure 10.19).

Organ of corti

The **organ of corti** (figure.10.19) is a sensory ridge located on the top of **the Basilar membrane** and it contains numerous hair cells that are arranged in four rows along the length of the basilar membrane. Protruding from the apical part of each hair cell is hair like structures known as **stereocilia.** During the conduction of sound wave, stereocilia makes a contact with the stiff gel membrane called **tectorial membrane,** a roof like structure overhanging the organ of corti throughout its length.

Mechanism of hearing

Sound waves entering the external auditory meatus fall on the tympanic membrane. This causes the ear drum to vibrate, and these vibrations are transmitted to the oval window through the three auditory ossicles. Since the tympanic membrane is 17-20 times larger than the oval window,